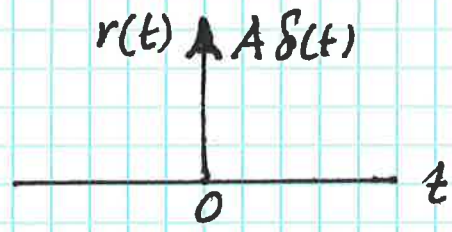
Chapter 5 - The Performance of Feedback Control System

5.1 Introduction

In this chapter we discuss time-domain performance specification and system response for specific input signals. First order, second order and higher order systems will be considered. System response due to location of system poles will be discussed. The system performance based on steady state error and system indices (measures) will be evaluated

5.2 Test Input Signals

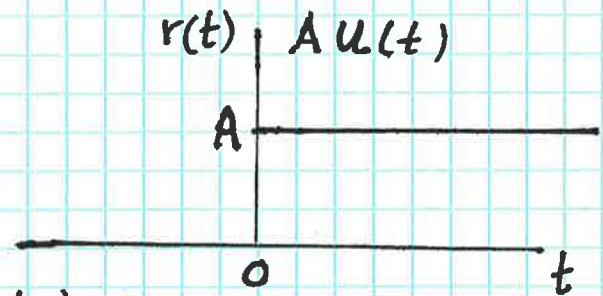


a) $r(t) = A\,\delta(t)$ impulse

$$= \infty \qquad t = 0$$
$$= 0 \qquad t \neq 0$$

$$\int_{-\infty}^{\infty} A\,\delta(t)\,dt = A \qquad \int_{-\infty}^{\infty} f(t)\,\delta(t-a)\,dt = f(a)$$

$$R(s) = \mathcal{L}[r(t)] = A$$



b) $r(t) = A\,u(t)$

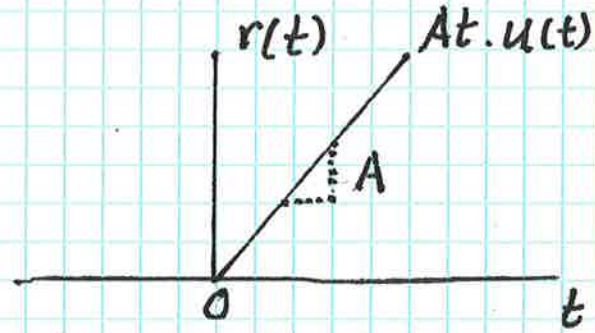
$$= 0 \qquad t < 0$$
$$= A \qquad t > 0$$

$A\,u(t) =$ step function
switching function

$$R(s) = \frac{A}{s}$$

c) $r(t) = At\,u(t)$

$$= 0 \qquad t < 0$$
$$= At \qquad t > 0$$

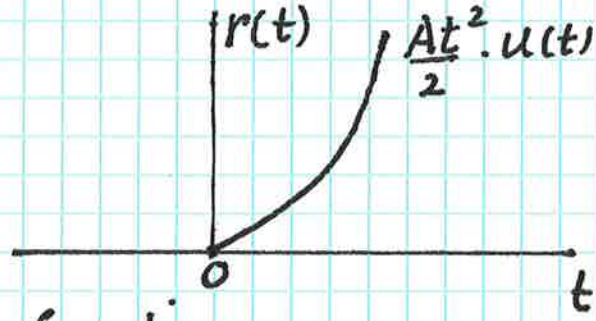


$At\,u(t)$ = ramp function

$$R(s) = \frac{A}{s^2}$$

d) $r(t) = \frac{A}{2} t^2 u(t)$

$$= 0 \qquad t < 0$$
$$= \frac{At^2}{2} \qquad t > 0$$

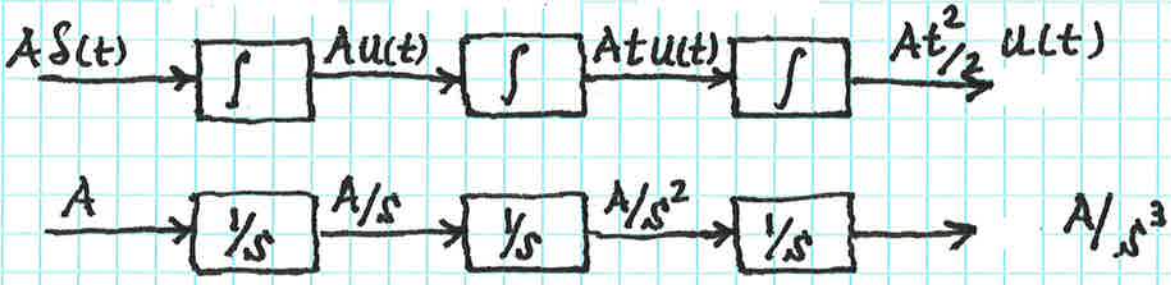


$\frac{At^2}{2} u(t)$ = parabolic function

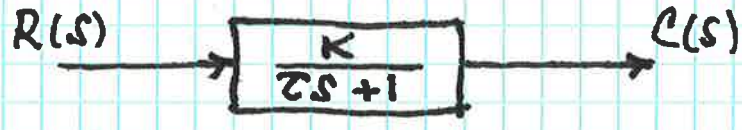
$$R(s) = \frac{A}{s^3}$$

$\vdots$

Note: $r(t) = t^n u(t) \Rightarrow R(s) = \frac{n!}{s^{n+1}}$



First Order System - Step & Impulse Response



Let $R(s) = \frac{1}{s}$ $\Rightarrow$

$$C(s) = \frac{K}{s(\tau s+1)} = \frac{K/\tau}{s(s+\frac{1}{\tau})} = \frac{A_1}{s} + \frac{A_2}{s+1/\tau}$$

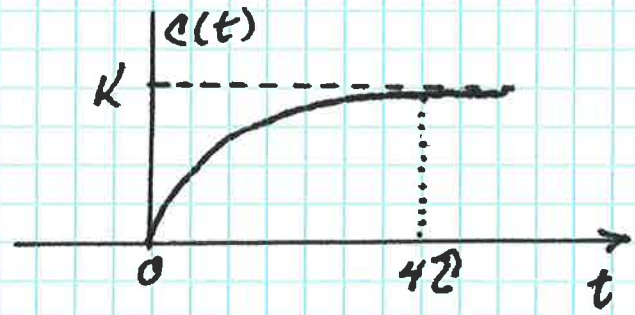
$$A_1 = K \quad , \quad A_2 = -K$$

$$c(t) = K - Ke^{-t/\tau} = K(1-e^{-t/\tau})$$

$C_{ss} = K$

T_s = Settling time $= 4\tau$

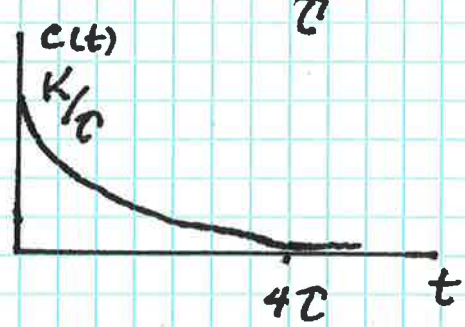
= time for $c(t)$ to reach 98% of final (steady state) value



Question: How does the results shown above change if input is $R(s) = \frac{A}{s}$?

Let $R(s) = 1$ $\Rightarrow$

$$C(s) = \frac{K}{\tau s+1} = \frac{K/\tau}{s+1/\tau} \Rightarrow c(t) = \frac{K}{\tau} e^{-t/\tau}$$



* Note: The derivative of step response, $c(t) = K(1-e^{-t/\tau})$ is impulse response, $c(t) = \frac{K}{\tau} e^{-t/\tau}$

This should also be clear by comparing $C(s) = \dfrac{K/\tau}{s(s+1/\tau)}$ for step input to $C(s) = \dfrac{K/\tau}{s+1/\tau}$ for impulse input.

Now consider step response for a the first order system with $c(0) = c_0 \neq 0$:

$$R(s) \longrightarrow \boxed{\frac{K}{\tau s+1}} \longrightarrow C(s)$$

$$\frac{C(s)}{R(s)} = \frac{K}{\tau s+1} \Rightarrow \tau \frac{dc}{dt} + c(t) = Kr(t)$$

$$\tau\left[s\,C(s) - c_0\right] + C(s) = K\left[\frac{1}{s}\right]$$

$$[s\tau+1]\,C(s) = \tau c_0 + K\left[\frac{1}{s}\right]$$

$$C(s) = \frac{\tau c_0}{\tau s+1} + \frac{K}{s(s\tau+1)} = \frac{c_0}{s+\frac{1}{\tau}} + \frac{\frac{K}{\tau}}{s(s+\frac{1}{\tau})}$$

$$c(t) = c_0 e^{-t/\tau} + K(1-e^{-t/\tau})$$

Note: There is no reason to memorize these equations.

a) If initial condition is zero, evaluate

$$C(s) = R(s)\,G(s),$$

and take Inverse Laplace.

b) If initial condition is not zero (c_0), convert transfer function $\frac{C(s)}{R(s)}$ to differential equation and take Laplace Transform of the differential equation remembering $\mathcal{L}\left[\frac{dc}{dt}\right] = sC(s) - c_0$

5.3 Performance of Second-Order Systems

Let $R(s)$ = input and $Y(s)$ = output.

Consider

$$R(s) \longrightarrow \boxed{\dfrac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}} \longrightarrow Y(s)$$

We saw: $s^2+2\zeta\omega_n s+\omega_n^2=0$

$$s=-\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$$

a) $\zeta>1$ Two negative real distinct roots
Overdamped

$$s_1=-\zeta\omega_n+\omega_n\sqrt{\zeta^2-1}$$
$$s_2=-\zeta\omega_n-\omega_n\sqrt{\zeta^2-1}$$

b) $\zeta=1$ Two negative real repeated roots
Critically Damped $s_1=-\omega_n$, $s_2=-\omega_n$

c) $0<\zeta<1$ Two complex roots with negative real part
Underdamped

$$s_1=-\zeta\omega_n+j\omega_n\sqrt{1-\zeta^2}$$
$$s_2=-\zeta\omega_n-j\omega_n\sqrt{1-\zeta^2}$$

d) $\zeta=0$ Two imaginary roots
Undamped $s_1=j\omega_n$, $s_2=-j\omega_n$

Step Response of an Underdamped System:

$$Y(s) = \frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)} \qquad 0<\zeta<1$$

$$= \frac{A_1}{s} + \text{portion due complex poles } -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$A_1 = \left.\frac{s\,\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}\right|_{s=0} = 1$$

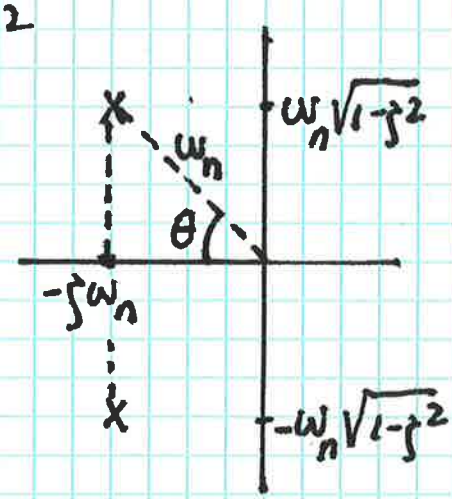
Using partial fraction method for complex poles (page 26 of lecture notes), we have

$$Q(s) = \left.\frac{\omega_n^2}{s}\right|_{s=-\zeta\omega_n+j\omega_n\sqrt{1-\zeta^2}}$$

$$= \frac{\omega_n^2}{-\zeta\omega_n+j\omega_n\sqrt{1-\zeta^2}}$$

$$= \frac{\omega_n^2}{\omega_n \angle -\tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}+\pi}$$

$$= \omega_n \angle \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}-\pi$$



Hence:

$$y(t) = 1 + \frac{\cancel{\omega_n}}{\cancel{\omega_n}\sqrt{1-\zeta^2}}\, e^{-\zeta\omega_n t} \sin\left(\omega_n\sqrt{1-\zeta^2}\, t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta} - \pi\right)$$

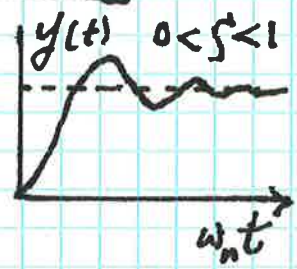
Let $\beta = \sqrt{1-\zeta^2}$ and note $\theta = \tan^{-1}\dfrac{\sqrt{1-\zeta^2}}{\zeta}$

$$= \tan^{-1}\frac{\beta}{\zeta}.$$

Also note $\sin(\alpha - \pi) = -\sin\alpha \implies$

$$\boxed{y(t) = 1 - \frac{1}{\beta}e^{-\zeta\omega_n t}\sin(\omega_n\sqrt{1-\zeta^2}\,t + \theta)}$$

Also note, $\theta = \cos^{-1}\dfrac{\zeta\omega_n}{\omega_n} = \cos^{-1}\zeta$



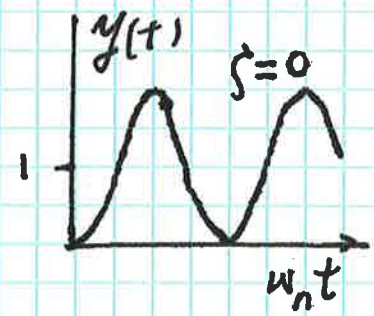
Step Response of an Undamped System:

$$Y(s) = \frac{\omega_n^2}{s(s^2+\omega_n^2)}, \qquad \zeta = 0$$

Since this case and Under damped are of the same family, instead of partial fractions (Case IV, page 27 of lecture notes), set $\zeta = 0$ in the above equation:

$$y(t) = 1 - \sin(\omega_n t + \theta), \qquad \theta = \cos^{-1}0 = \frac{\pi}{2}$$

$$y(t) = 1 - \sin\left(\omega_n t + \frac{\pi}{2}\right) = 1 - \cos\omega_n t$$



Step Response of an Overdamped System :

$$Y(s) = \frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)} \qquad \zeta > 1$$

Poles : $s = 0,\ -\zeta\omega_n + \omega_n\sqrt{\zeta^2-1},\ -\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}$

Take Inverse Laplace. Solution will be of the form;

$$y(t) = 1 + A_2 e^{(-\zeta\omega_n+\omega_n\sqrt{\zeta^2-1})t} + A_3 e^{(-\zeta\omega_n-\omega_n\sqrt{\zeta^2-1})t}$$

Step Response of a Critically Damped System :

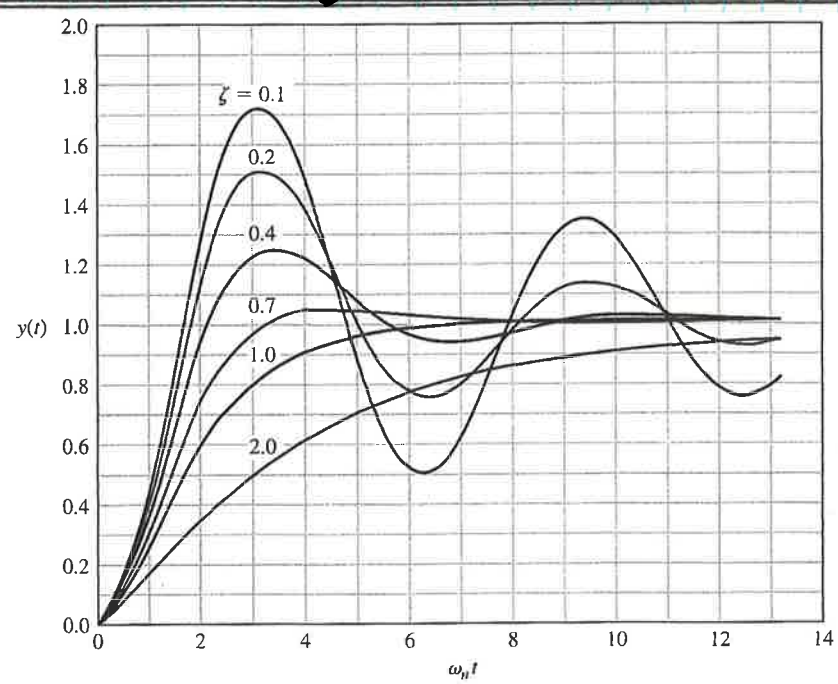
$$Y(s) = \frac{\omega_n^2}{s(s^2+2\omega_n s+\omega_n^2)} = \frac{\omega_n^2}{s(s+\omega_n)^2}, \quad \zeta = 1$$

Poles : $s = 0,\ -\omega_n,\ -\omega_n$

Take Inverse Laplace. Solution will be of the form;

$$y(t) = 1 + A_2 e^{-\omega_n t} + A_3 t e^{-\omega_n t}$$

Here are step response plots of some underdamped, overdamped and critically damped. For step response of undamped ($\zeta = 0$), see the previous page.



Impulse Response of an Underdamped System:

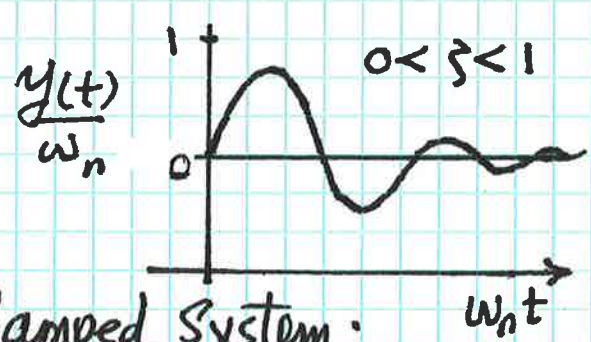
$$Y(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$0 < \zeta < 1$$

$$Q(s) = \omega_n^2 \Big|_{s = -\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2}} = \omega_n^2 \angle 0^\circ$$

$$y(t) = \frac{\omega_n^2}{\omega_n\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin\left(\omega_n\sqrt{1-\zeta^2} + 0^\circ\right)$$

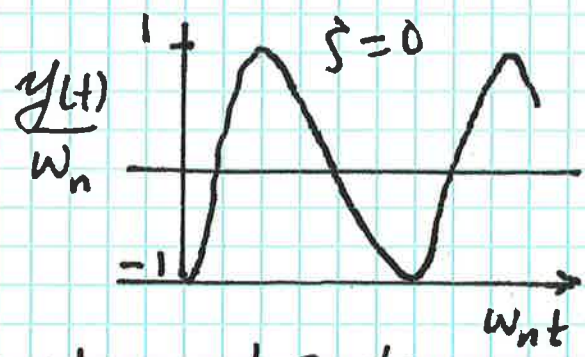
$$\boxed{y(t) = \frac{\omega_n}{\beta} e^{-\zeta\omega_n t} \sin(\omega_n \beta t)}$$



Impulse Response of an Undamped System:

Set $\zeta = 0$ in the above equation:

$$y(t) = \omega_n \sin \omega_n t$$



Impulse Response of an overdamped System:

$$Y(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad \zeta > 1 \qquad \text{Poles: } -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Take Inverse Laplace. Solution will be of the form

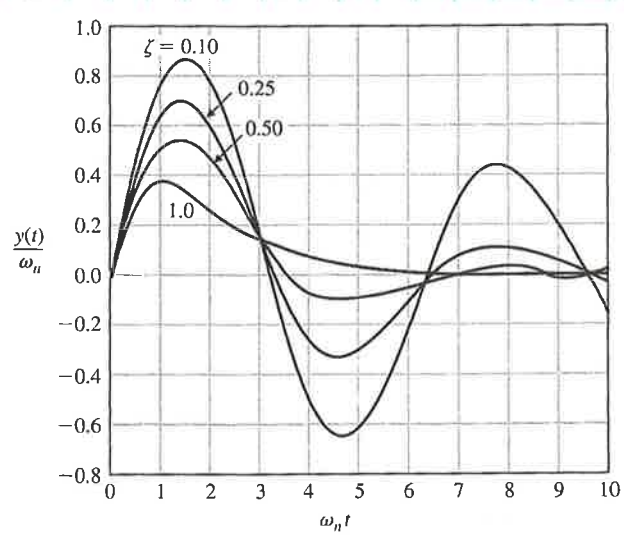
$$y(t) = A_1 e^{(-\zeta\omega_n + \omega_n\sqrt{\zeta^2-1})t} + A_2 e^{(-\zeta\omega_n - \omega_n\sqrt{\zeta^2-1})t}$$

Impulse Response of a Critically Damped System:

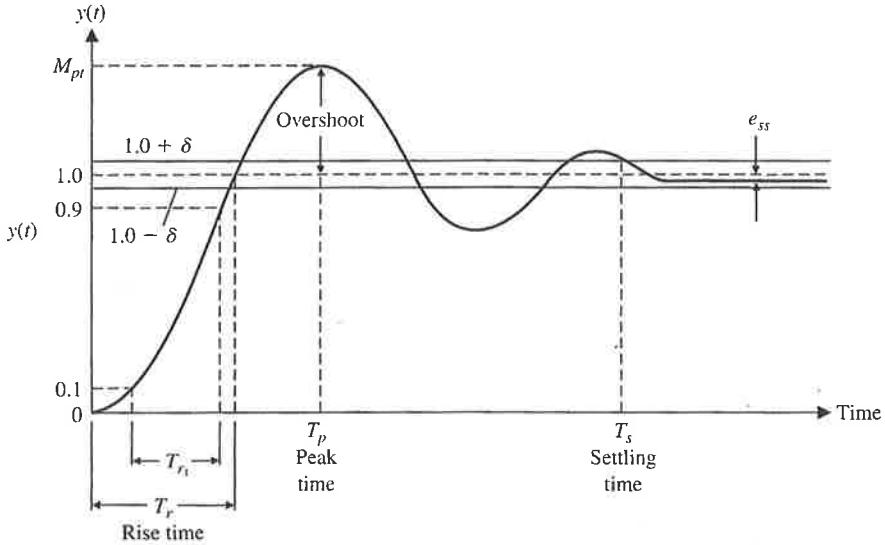
$$Y(s) = \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s+\omega_n)^2}$$

$$y(t) = \omega_n^2 t\, e^{-\omega_n t}$$

Here are impulse response plots of some underdamped, overdamped and critically damped. For impulse response of undamped ($\zeta = 0$), see the previous page.



Now let us consider step response of an underdamped system ($0 < \zeta < 1$):



Peak Response M_{pt} : Maximum value of response denoted by M_{pt}.

T_p : Time to reach maximum response.

f_v or y_{ss} : Steady state value or final value of response.

P.O. : percent overshoot.

T_r : Rise time, time for the response to rise from 10% to the final value of 90% of the final value.

T_s : Settling time, time required for the output to settle within a certain percent of the final value.

$$c(t) = 1 - \frac{1}{\beta} e^{-\zeta \omega_n t} \sin\left(\omega_n \sqrt{1-\zeta^2}\, t + \theta\right)$$ step response $0 < \zeta < 1$

T_p:

Take the derivative of step response $c(t)$ and set equal to zero. Note the derivative of step response is the impulse response which was obtained as:

$$\frac{\omega_n}{\beta} e^{-\zeta\omega_n t} \sin \beta\omega_n t = 0$$

$$\beta\omega_n t = \pi \implies \boxed{T_p = \frac{\pi}{\beta\omega_n} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}}$$

M_{pt}:

Substitute $t = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}$ in step response to obtain the maximum value.

$$(c(t))_{max} = M_{pt} = 1 - \frac{1}{\beta} e^{-\zeta\omega_n\left[\frac{\pi}{\omega_n\sqrt{1-\zeta^2}}\right]} \cdot \sin\left(\beta\omega_n \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} + \theta\right)$$

$$M_{pt} = 1 - \frac{1}{\beta} e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}\left[-\sin\theta\right]$$

$$M_{pt} = 1 + \frac{1}{\cancel{\beta}} e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \cdot \frac{\cancel{\beta}}{1}$$

$$\implies \boxed{M_{pt} = 1 + e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}}$$

P.O.:

$$P.O. = \frac{M_{pt} - f_v}{f_v} = \frac{1 + e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} - 1}{1}$$

$$\boxed{P.O. = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100\%}$$

f_v or y_{ss}:

$$f_v = y_{ss} = \lim_{t \to \infty} y(t) = \lim_{s \to 0} sY(s) = 1$$

T_s :

$$T_s = 4\tau = 4\left[\frac{1}{\zeta\omega_n}\right] = \frac{4}{\zeta\omega_n}$$

T_r : Rise Time

Rise time is measured from response. Furthermore T_p provides similar information as T_r of how fast the response is moving.

Note: The equations for P.O. and T_p are used to plot the figure 5.8 that can be used to obtain T_p & P.O. for different values of damping ζ.

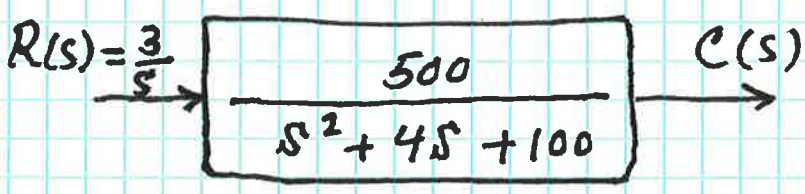
Example. For the system shown, evaluate:

a) ω_n, ζ, $\omega_d = \omega_n\sqrt{1-\zeta^2}$

b) T_s

c) M_{pt}

d) T_p

e) $f_v = y_{ss}$

f) P.O.